Intro to Web Dev

Part 1



x



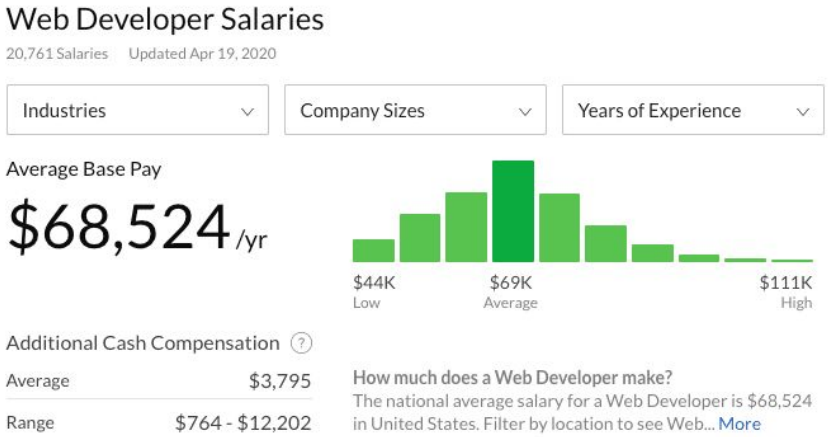






Web Development ?

- Create websites
- Websites can be super complex (ex: Amazon. Netflix, Canvas)
- High demanding job opportunities, great pay



Frontend, Backend, Full-stack

Website development can be divided into **3 categories**: frontend, backend, full-stack

- **Frontend**: work on the user interface of the website (the beauty, the aesthetic side)

- **Backend**: work on the logic behind website (interact with database, authorize users, transactions,...)

- **Full-stack**: Frontend + Backend



Intro to Web Dev Series

What is it about?

- Technical series that will demonstrate all the skills needed to create a fully working website and best coding practices when it comes to web dev

Goal of the series

- Able to create a functional website project from scratch and deploy it

Contents of the series?

- Web Dev 1 (HTML / CSS)
- Web Dev 2 (JavaScript - Svelte)
- … coming soon

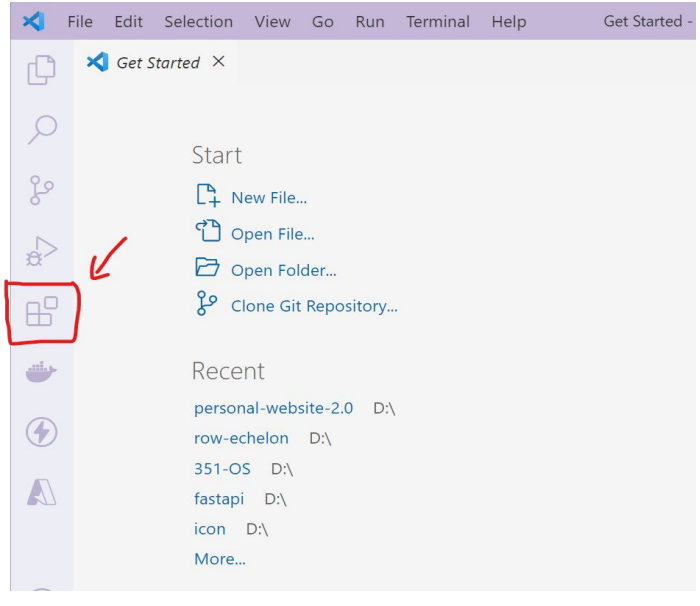
Website Template Explained

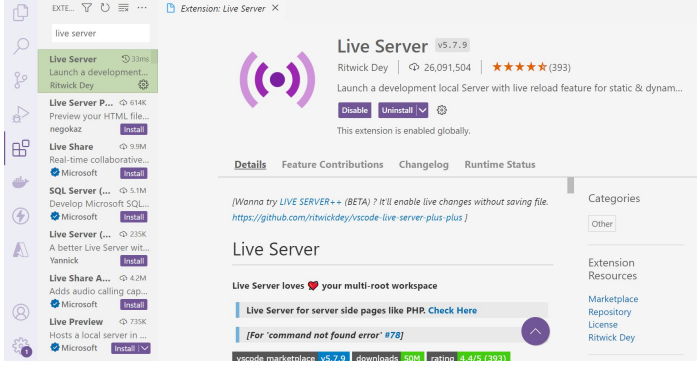
We’ve cloned the template, what now?

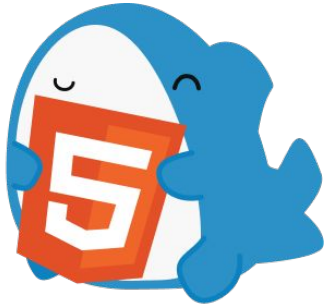
- Svelte uses HTML and CSS so it’ll be beneficial to learn the fundamentals first by practicing them
- What we work on during this session will be very easy to transfer into a Svelte file later!

Before We Start

- Install VSCode (Text Editor - where you will code)
- Install Live Server from VSCode Extensions (help you preview your local website)







HTML Basics

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HTML (HyperText Markup Language)

What is it?

- The very basic markup language to help you build the skeleton of your website

Get Started

- Create an index.html file in VSCode
- In the file, type html:5 and press Enter for autocomplete

index.html > ...

```
1   <!DOCTYPE html>
2   <html lang="en">
3   <head>
4       <meta charset="UTF-8">
5       <meta http-equiv="X-UA-Compatible" content="IE=edge">
6       <meta name="viewport" content="width=device-width, initial-scale=1.0">
7       <title>Document</title>
8   </head>
9   <body>
10
11  </body>
12  </html>
```

<- It'll look like this as the result

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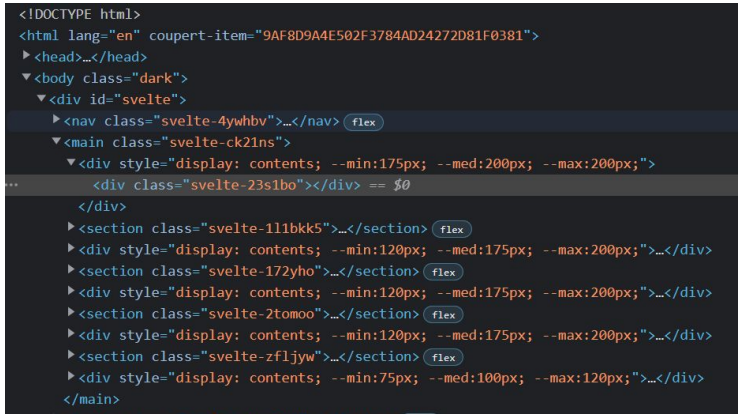
HTML – Tags <> </>

What is it?

- In HTML we use **tag** to define website elements, syntax is <open tag> </close tag> (Ex: <img> for image </img>, for paragraph),

Tags are everywhere

- Go to any website > right click > inspect elements



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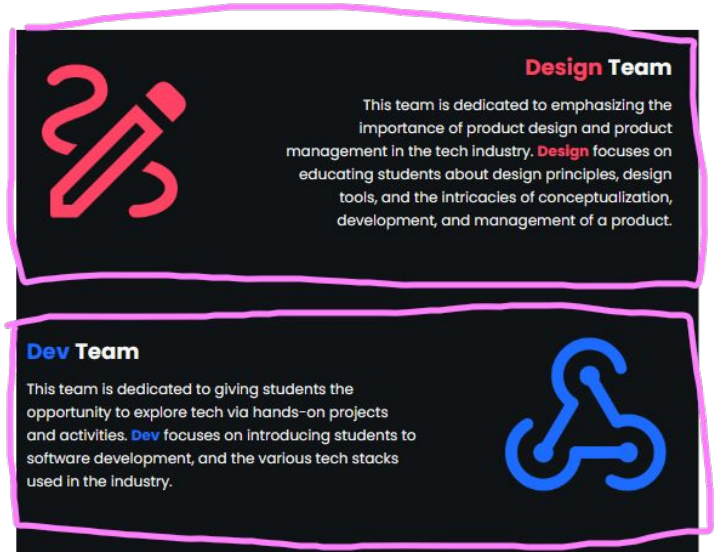
Common tags

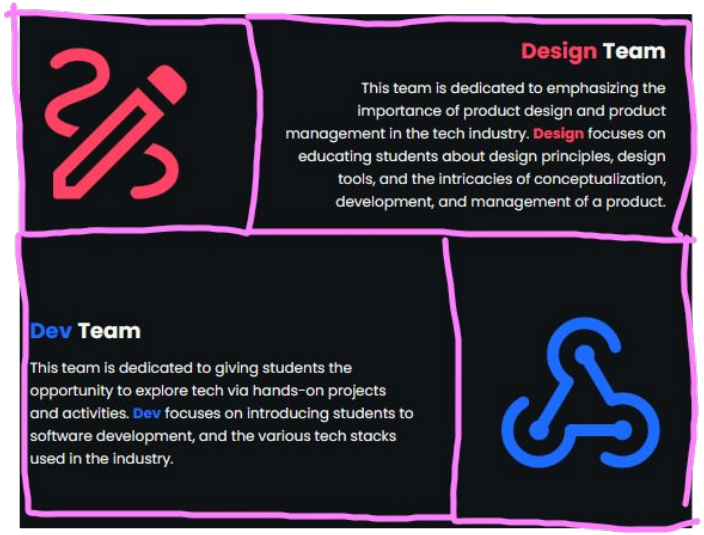
- **<title> </title>** : Website title (show on navigation bar)
- **<body> </body>** : Defines body elements (should be wrap around all elements)
- **<h1> </h1>**: Website most important heading (heading size/importance order h1 -> h6)
- ** **: Defines a paragraph
- ** **: Defines division/block/container of element
- **<img src=”your image link.jpg” alt=”text when hover”> </img>:** Defines image
- **link**: Defines hyperlink

The power of

- Understand how   works = Understand the website layout

  Example:





Ordered and Unordered Lists

<ol> and **<ul>** are tags used to create lists of items. Ordered lists are numbered while unordered lists are just bullet points. **<li>** tags are used to define the contents of the lists.

Example:

```
<ol>
    <li>Ordered</li>
    <li>List</li>
</ol>

<ul>
    <li>Unordered</li>
    <li>List</li>
</ul>
```

1. Ordered
2. List

- Unordered
- List

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Images

We've gotten some different text tags down, but what if we want a picture? We can use the <img> tag to reference them. We also include an alt in case the image can't load, so people know what should have been there

Example:

```
<img src="https://www.fullerton.edu/webstandards/_resources/images/campus_stock/CSUF_dusk.jpg" alt="CSUF Pic">
```



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HTML Challenge Time

- Make a **html file** and name it whatever you want (ex: danielsohot.html)
- Make the **title** of the website your name
- Display a **BIG** cute greeting message
- Display a **not so big** message under greeting message
- Display an **image** of whatever you like (ex: Thug Frank)
- Insert your favorite **website link** such that user will be redirected to that website when they click on your link
- Create a **bullet-point list** includes anything (facts about yourself, hobbies,...)
- **BONUS**: An image with link that when user click on the image, they'll be directed to the attached link

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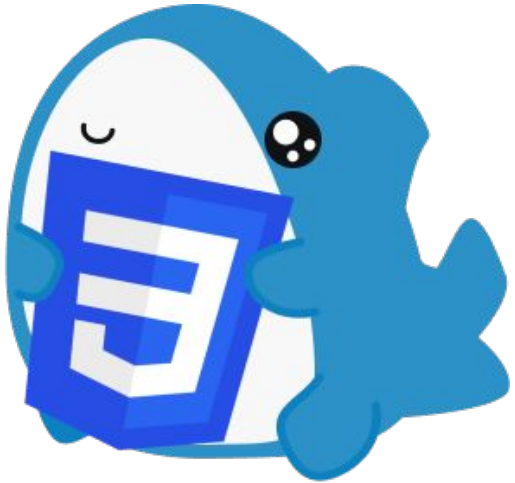
CSS Basics

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What is CSS?

- ➔ Cascading Style Sheets
- ➔ The skin of a website

CSS uses selectors to specify which element(s) to style + props to declare values



```
selector {
    prop: value;
}
```

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How to Select Elements

Elements can be selected 3 ways: by tag, by class name, by id

// html

```
<h1>blah blah blah</h1>
<h2 class="blah">blah blah blah</h2>
<h3 id="uniqueBlah">blah blah blah</h3>
```

// css

```
h1 { color: red; }

.blah { color: red; }

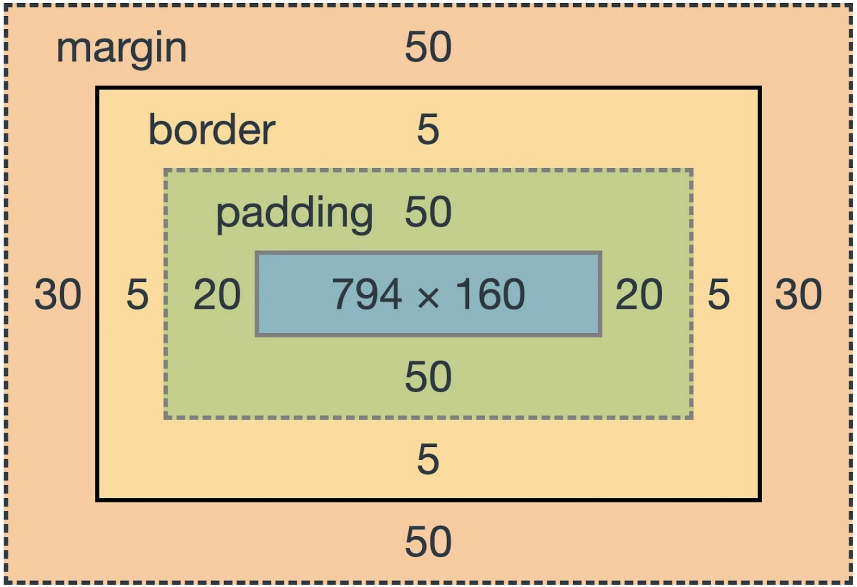
#uniqueBlah { color: red; }
```



The Box Model

- ➔ Style the spacing around a element you target

- ➔ Blue: Height x Width of element
- ➔ Green: Spacing inside of container
- ➔ Yellow: Optional border sizing
- ➔ Orange: spacing outside of container



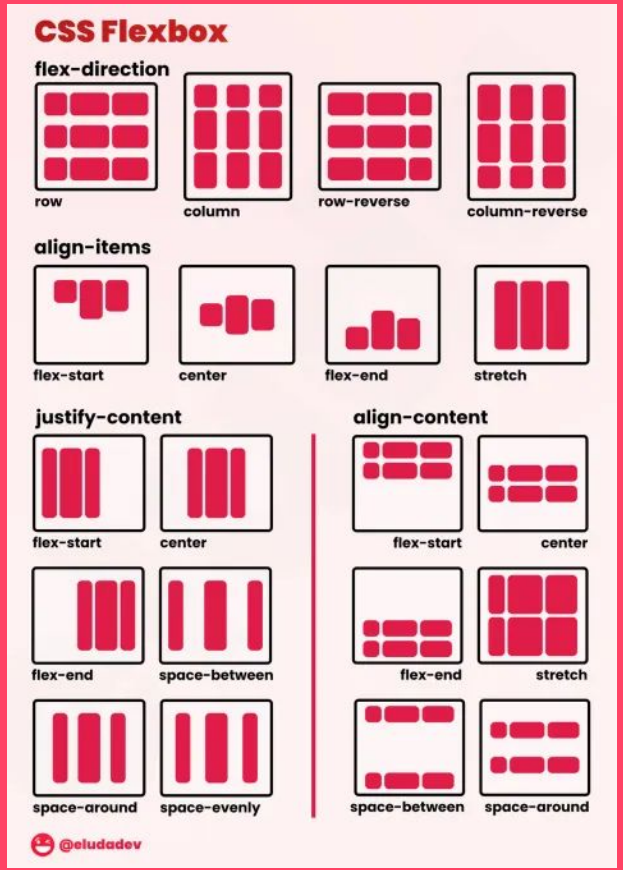
Visualize Example

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Flexbox

This is a crazy positioning tool as hell and I can not cover it in 5 minutes 😔

- ➔ To the right is a graph of a summary on how flexbox can organize content in a container 😮
- ➔ Click here for a game to understand flexbox 🎮



More Cool Properties

- ➔ **margin:** [units];
- ➔ **padding:** [units];
- ➔ **width:** [units];
- ➔ **height:** [units];
- ➔ **border:** [color] [thickness] [style];
- ➔ **border-radius:** [units];
- ➔ **text-align:** [position];
- ➔ **box-shadow:** [x-axis] [y-axis] [blur in units] [color];
- ➔ **color:** [color];
- ➔ **background-color:** [color];
- ➔ **font-family:** [font family];
- ➔ **font-size:** [units];
- ➔ **font-style:** [italic / bold / etc];
- ➔ **font-weight:** [units in base 100];
- ➔ **font-shadow:** [x-axis] [y-axis] [blur in units] [color];

and more...

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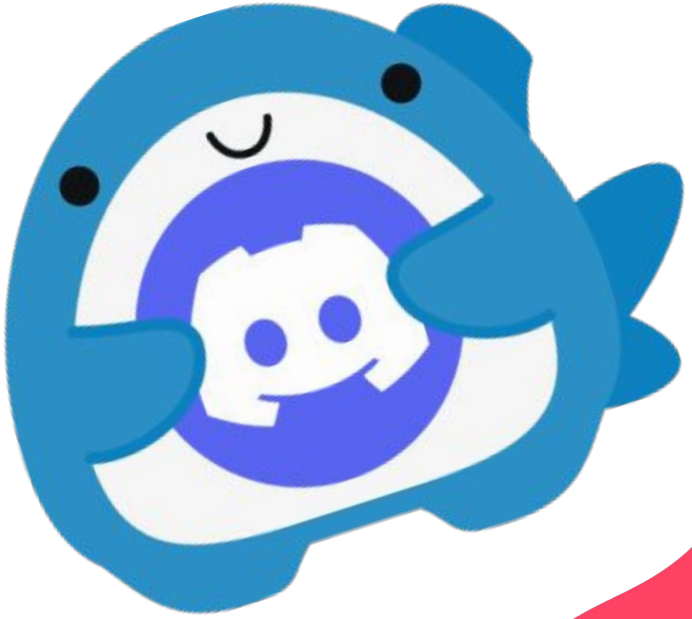
CSS Challenge Time

- Make a **style tag** inside of your <head> tag ✨
- **apply classes or id's** inside of the html tags *(slide 17)* 🆔
- target html **classes, id's, or html** tags inside of the style tag *(slide 17)* 🎯
- **apply properties** inside of the targeted element *(slide 18-20)* 🎉
- play around with spacing like **margin or padding** *(slide 18)* 🌌
- **BONUS**: adjust the position of the elements using flexbox *(slide 19)* 🤯

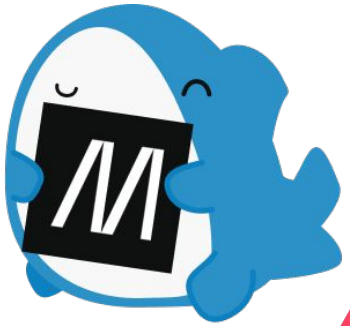
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CSS Resources

- ➔ Kevin Powell on Youtube
- ➔ Box Model Docs
- ➔ MDN CSS Docs
- ➔ Flexbox Guide
- ➔ #design-team *or #dev-team*







Reminder

Please register on Discord to access all channels

